

Andrea Yang

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Research/Academic Experience

Genomics @ Georgetown University / Summer Science Program (SSP)

Washington D.C. · Summer 2024

Won individual \$1K stipend for stress response research in *Vibrio natriegens* with erythromycin in relationship to genetics and the bacterial growth curve in 6 weeks. [Poster](#) and manuscript available upon request.

Karginov Lab @ UCR

Riverside, CA · August 2024-August 2025

Developing neural ODEs for forecasting glioma growth. Publishing *Pseudomonas aeruginosa* antibiotic resistance mechanisms and AI therapeutics review paper at ACS Omega, review status pending; manuscript available upon request. Published independent research developing an automated R-based AlphaFold 3 pipeline to predict protein-protein and ion interactions in *Mus musculus*. Streamlined iPTM/pTM data analysis revealed significant Endo-U interactions, advancing computational frameworks for RNA regulation and antibiotic resistance.

Balazsi Lab @ Stony Brook University

Remote · March 2024-Present

Modeled bistability in gene regulatory circuits to investigate causes of bimodal gene expression underlying disease states; independently derived and simulated first-order kinetic, ODE, and Hill models in Python and MATLAB under mentorship from Dr. Balazsi to characterize conditions driving cellular state transitions. Presented code + equations in research talk to faculty.

Bogdan Lab @ USC

Los Angeles, CA · June 2024-Present

Developed dual Fourier Neural Operator architecture to solve PDE inverse problems. Integrated latent-space coupling and explored causality to enhance accuracy. Created [radar plots](#) for computational neuroscience project. Contributed literature review and experiments to L4DC conference submission.

Work/Jobs

Starfly / Founder

Aug 2022-Present

Founder of Starfly, an iOS educational app nonprofit 501(c)(3) in the state of California for helping the gender gap in STEM fields. Coded with Swift in Xcode: employs advanced AI/ML educational models, as well as augmented reality (ARKit), [demonstration link](#) here.

- Implemented into local educational groups and clubs. Did all programming, graphic design (UX/UI), pilot testing, outreach, publicity. Partnered with University of Southern California Roski, Rise at Schmidt Futures. Received press coverage + winner of 2024 Congressional App Challenge + named **Rise For The World Global Finalist** (<10% acc. rate challenge for global talent)

Hack Club / Community Team, Organizer, Leader

Jul 2022-Present

Co-lead organizer of [Ascend](#), Hack Club Athena's initial summit to bring together girls and women in tech; partnered with SpaceX, raised \$50K+. **Led epsilon-greedy/Q-learning workshop at SpaceX Hawthorne**. Co-lead organizer of HackOC (<https://hackoc.org>). Hackathon sponsored by Anduril, First, Hack Club, and more. Raised \$20K+, and saw 100+ high school attendees from all over Southern California. Co-lead organizer of Halo (raised 15K+, <https://halo.hackclub.com>). Days of Service event (<https://days-of-service.hackclub.dev/>) with Hack Club; Saw over 50 young girls attend. Halo dinner invited women that were SpaceX engineers, millionaires in STEM, game developers at Riot, and more out to a multi-million mansion in Los Angeles as a networking event. Lead organizer of [Odyssey](#), Los Angeles' first high school, girls-only AI hackathon, partnered w/ Snapchat. Also: club leader @ HC.

- Apocalypse hackathon project, an Arduino home-alarm system, featured in GitHub Universe 2024.

Neuronex / Team Lead

Aug 2024-Present

Led investigation of ultrasound and ML models applied to diagnosing neurodegenerative disease. Developed complete sonography circuit and high ML algorithm accuracy. Engineered computing pipeline from scratch. **Conrad Challenge Innovator** status and garnered grant interest from Bagel Fund.

Awards/Accolades

- 2x USA Biology Olympiad (USABO) National Semifinalist (first freshman semifinalist at Diamond Bar High School)
- 2024 USABO National Top 150 (top 1%)
- NCWIT National Honorable Mention + LA Affiliate Award
- Biology Bowl National Semifinalist
- SSP Chambers Scholar (0.4% rate overall selected for scientific excellence)
- AIME Qualifier
- 2x Gold Presidential Service & Volunteer Award
- Piano Certificate of Merit Level 10/Advanced w/ State Honors
- BuildGirls Scholar (\$2.5K)
- AwesomeMath Scholar
- 2025 Coolidge Senator
- CA38 Congressional App Challenge Winner (invited to visit US Capitol Building to present app to members of Congress, attend STEM Expo, featured on House of Reps site and #HouseofCode Festival)
- [U.N. Report Publication](#) for rare disease advocacy

Education

B.S. Electrical Engineering & Computing @ Massachusetts Institute of Technology (MIT)

Cambridge, MA · *Expected grad: May 2030*

UW GPA: 5.0/5.0. Minor in philosophy

Relevant coursework: 6.1010 Fundamentals of Programming, 6.100A Introduction to CS Programming in Python, 7.01x Introductory Biology, 18.02 Multivariable Calculus, 8.02 Physics II: Electricity & Magnetism, 21.01 Compass Course: Moral and Social Questions about the Human Condition

Diamond Bar High School

Diamond Bar, CA · *Aug 2022 - May 2026*

UW GPA: 4.0/4.0, class rank 1/651.

Relevant coursework (all scores 5/5): AP Calculus BC, AP Calculus AB, AP European History, AP U.S. History, AP English Language/Composition, AP Chemistry, AP Physics 1, AP Computer Science A, AP Biology (self-study). Currently taking: AP Physics C: Mechanics, AP Physics C: E&M, AP Spanish Language & Culture, AP Eng Literature, AP Microeconomics, AP Statistics.

Dual enrollment at College of Siskiyous for Multivariable Calculus (Calculus III). Grade: A/A

Clubs/activities: Science Olympiad (8+ years), medaled at the state + national invitational level at Caltech, USC, UCR, etc. in biology and chemistry. Cumulative 40+ medals. USABO Club Co-President, Code Club President, Science Bowl Biology Lead.
